Measured Self-Occlusion in a Deployed Cross-Sensor Diagnostic Layer: When a Fault Destroys the Evidence That Would Name It

Andres Felipe Pena Avila
SWORT LLC, Bethesda, Maryland, United States

Abstract

We report measurements of *false-healthy exposure* on a deployed cross-sensor diagnostic layer for small uncrewed aircraft: the time the system spends asserting health while holding no evidence it is entitled to assert it from. The layer maintains per-sensor trust coefficients over a physics-derived graph and runs a gated naive-Bayes diagnosis over eight mutually exclusive causes. A gated feature is skipped rather than down-weighted, and the implementation has no state meaning "not assessable", so absence of applicable evidence is emitted as health. Under sustained GNSS denial the diagnosis reports `nominal` for the entire fault window and never names `gps_fault`, while the GPS trust coefficient reads 0.9767. The vehicle is not blind: the failsafe reaches CAUTION through an independent data-freshness disjunct and reconfigures navigation. It cannot name the cause.

Two scope corrections, both forced by our own controlled-injection campaign against a real autopilot, are reported here rather than buried. First, this behaviour is specific to faults that remove the GNSS fix. A position glitch that leaves the fix valid does not close the gate at all, so the evidence remains available. Second, in that case the layer sees the fault and still does not escalate: its residual detector saturates at the floor for every injected position error we tested, from 5.6 m to 5.6 km, and no step magnitude produces an alarm. The same total error applied as a ramp does. Escalation here is duration-limited, not magnitude-limited.

We make no observability claim. Receiver observables detect GNSS denial well; the failure we measure is one of evidence routing and applicability semantics, not of detectability. We report a threshold result that yields a checkable design rule, a chatter-resonance prediction tested against the deployed code and holding, a one-line repair with its full measured cost, and a counterfactual abstaining policy measured against current behaviour on the same flights. We claim engineering contribution and measurement only.

1 Introduction

Model-based diagnosis on a real vehicle rarely evaluates every residual at every instant. Residuals are gated: a cross-check between barometric and GNSS altitude is meaningless without a GNSS fix, and a thrust-versus-current relation is meaningless before the motors are armed. Gating is correct engineering. It is also where a specific failure mode lives.

The failure mode is this. If a fault closes a gate, and the features that gate controls are the only evidence for that fault, then the fault removes its own evidence. Nothing in the diagnosis reports an anomaly, because the anomalous feature is not evaluated at all. The system does not become uncertain. It becomes confident, and wrong, in the direction of health.

We measured it. On the deployed layer, with a GPS denial fault active for 40 s, the diagnosis reported `nominal` on 100% of post-onset ticks and reached `gps_fault` zero times. The GPS trust coefficient over the same window never fell below 0.9767, because a channel that stops delivering data has its belief update frozen rather than driven down: absence of evidence is not treated as

evidence of fault. Both the attribution channel and the trust channel independently reported health while a fault the layer was built to catch was active.

The vehicle is not blind to the condition, and we state that before any reader can form a wrong impression. The failsafe policy escalates to CAUTION and issues `SWITCH_NAV_INERTIAL` and `INHIBIT_GPS_MODES` in this scenario, driven by a data-freshness disjunct that does not consult the diagnosis. The gap is in attribution, not in escalation. What is lost is the ability to name the cause, and therefore the ability to choose a cause-specific response, to log a correct incident record, or to distinguish GNSS denial from a benign loss of reception.

The condition is narrower than the failure mode described above, and we bound it here rather than in the limitations. A controlled-injection campaign against a full ArduPilot autopilot (Section 6) finds that faults which remove the GNSS fix close the gate almost always, and that a 62 to 76 m position glitch which keeps the fix does not close it at all. Self-occlusion is a property of fix-loss faults, not of GNSS faults in general.

We make no claim that GNSS denial is hard to observe, because it is not. Receiver-level observables detect it: carrier-to-noise density, automatic gain control behaviour, loss of lock, satellite count, correlation-peak distortion and RAIM status are all established means, and none of them is exotic. The failure we report is that this system's diagnosis is routed entirely through position-domain relations that its own validity gate switches off, and that it has no vocabulary for "I cannot assess this". It is an evidence-routing and applicability-semantics failure, and Section 12 prices the fix at one ungated relation.

We make no mathematical claim. Diagnosis under intermittent and permanent loss of observations is established [**?**, **?**], as is diagnosability under state-dependent observation [**?**] and, for the fault-state-dependent case that ours is an instance of, [**?**]. Structural isolability with mode-dependent signatures is established [**?**, **?**], and the reliability-side machinery we apply is likewise established [**?**, **?**].

Contribution. We studied which GNSS fault classes a deployed cross-sensor diagnostic layer detects, which it escalates, and how long it reports health without applicable evidence, using controlled injection against a real autopilot; we report the fault classes where it and the vehicle's own estimator are both blind, a closed-form account of why detection does not imply escalation, and the fault-injection harness we used.

Fault taxonomy. Every result below is reported against these four classes, distinguished by what each does to *applicability*: whether the test for `gps_fault` can be evaluated at all.

Class	Mechanism	Applicability	Consequence
fix-loss denial	receiver reports no 3D fix	$A_h \to 0$	evidence destroyed; self-occlusion
receiver link loss	serial corruption, fix lost	$A_h \to 0$	as above, by a different route
retained-fix step	position jumps, fix valid	$A_h = 1$	evidence intact; detected, not escalated
retained-fix ramp	position drifts, fix valid	$A_h = 1$	evidence intact; escalated only if slow and large enough

Table 1: Fault taxonomy. The applicability column is what separates self-occlusion from every other failure in this paper.

Harness. The injector and its all-layer recorder are the reusable part of this work and are prepared for release alongside the paper. They drive a real autopilot in software-in-the-loop, inject a parameterised GNSS fault, and record the receiver output, the estimator's innovation test ratios, every diagnostic residual before smoothing, the trust recursion before and after

update, the gate state and the final verdict, all on one timebase. Ground truth is the injection schedule, which is what makes the false-healthy quantities of Section 9 measurable at all.

2 Related work: what we instantiate

We list, for each framework, what we take from it and what we do not claim.

Diagnosis under unreliable observation. Carvalho et al. treat diagnosability when observations are lost intermittently [**?**] and permanently [**?**], including the case where the diagnoser is unaware that a sensor has failed. *What we take*: the framing that observation availability is part of the diagnosis problem rather than a precondition of it. *What we claim*: nothing. Our gate closure is an instance of observation loss in their sense.

State-dependent observation maps. Yin et al. develop diagnosability when the observation map depends on system state [**?**]. *What we take*: the recognition that a hypothesis-dependent observation map is a studied object. *What we claim*: nothing.

Fault-state-dependent observation loss. Dong, Yin and Li verify diagnosability under a state-dependent observation map $O : Q \times \Sigma \to 2^{\Delta \cup \{\varepsilon\}}$, which permits the observation available at a transition to depend on the state the system is in, including a fault state [**?**]. Our situation maps onto theirs directly: a fault-labelled augmented state corresponds to the hypothesis being active, $O(q, \sigma) = \{\varepsilon\}$ corresponds to gate closure silencing a feature, and failure of φ-diagnosability corresponds to what we call effective undetectability. *What we take*: the general mechanism of fault-induced observation suppression, entire. *What we claim*: nothing. Priority for general fault-induced suppression belongs to them. What is left to us is a measurement on a deployed system and a scoped enumeration of one system's evidence set, neither of which is a statement about the general mechanism.

Structural isolability and mode-dependent signatures. Krysander and colleagues compute isolability from structural models and select residuals that preserve diagnosability across operating modes [**?**], and compute minimal sensor sets for an isolability specification [**?**]. *What we take*: the design-time signature matrix and its induced partition, which we use as the baseline our measurement is compared against. *What we claim*: nothing. Our design-time analysis is theirs.

Fault trees and sensor selection by diagnostic performance. Hurdle et al. apply fault-tree methods to diagnosing system operation [**?**]; Reeves et al. select sensors against a diagnostic performance metric [**?**]. *What we take*: the whole apparatus of Section 12's cut-set analysis, unchanged. *What we claim*: nothing. Applying coherent-system structure functions to diagnostic capability is their contribution, not ours.

Isolation to the granularity of reconfiguration. Blanke et al. establish that faults need only be distinguished to the granularity that the reconfiguration map requires [**?**]. We use this to define the decision-relevant pair set in Section 3. *What we claim*: nothing.

3 System under measurement

The layer maintains a trust coefficient $T_i \in (0, 1)$ per sensor node over a graph of 8 nodes and 11 physics-derived edges. Each edge and each per-node physics self-model produces a chi-square-

normalised residual λ, mapped to a consistency score

$$c = \exp\!\big(-\tfrac{1}{2}\max(0, \lambda - 1)\big)\,, \tag{1}$$

so that only excess over the chi-square nominal penalises trust. Fused evidence updates a Beta belief with adaptive forgetting $\rho \in [0.95, 0.998]$, and the belief update is skipped entirely when a channel's data is stale.

Diagnosis is a gated naive Bayes over 8 mutually exclusive hypotheses. Each feature f carries a gate $g_f \in [0, 1]$, and the accumulation is

$$\ell_h \mathrel{+}= g_f \cdot \begin{cases} \log(1 - c_f + \varepsilon) & f \in \mathrm{LOW}(h) \\ \log(c_f + \varepsilon) & \text{otherwise,} \end{cases} \qquad \varepsilon = 0.02, \tag{2}$$

followed by exponential smoothing with $\alpha_s = 0.06$, tempering by 1.5, and a softmax against a prior placing 0.86 on `nominal`.

We call the output a normalized hypothesis weight, not a posterior. The three operations just listed each break the Bayesian reading, and any one of them alone would suffice: the evidence term is an exponential moving average over roughly fifteen ticks rather than the current tick's likelihood, it is raised to the power 1.5 before normalization, and $\varepsilon = 0.02$ is a floor inside the logarithm rather than a density. The weights sum to one and rank the hypotheses, which is what the assurance layer uses them for, but they are not $P(h \mid \text{observation at tick } k)$ and we do not report them as probabilities. Both the batch and the real-time paths compute the same quantity by the same three steps.

The load-bearing implementation detail is that a feature with $g_f \le 10^{-9}$ is skipped by a `continue`, not down-weighted. It therefore contributes to no hypothesis. Two hypotheses whose LOW sets coincide over the surviving features accumulate identical likelihood indefinitely, and the ratio of their normalized hypothesis weights remains pinned at the prior ratio. This is indistinguishability, not weak evidence.

3.1 Gates

Fourteen gates exist. Eight are data-freshness gates that block a node's belief update rather than any named feature. Two are compile-time constants that never actuate. Four vary with vehicle state and control a feature that some hypothesis references:

Gate	Predicate	Controls	Closed by
`armed_gate`	arming state, 1 s ramp	2 motor edges	nothing
`gvalid`	GNSS fix valid	5 features incl. `phys_gps`	`gps_fault`
`speed_gate`	ground speed 2–3 m/s	`heading_course`	undetermined
`gps_warmup`	elapsed time > 3 s	`phys_gps`	nothing

Table 2: The four actuating gates. Closure attribution is derived from source and the fault model; 29 of 32 hypothesis-gate cells are so derived, the three undetermined cells all concern `speed_gate` and none of them changes any partition reported here.

3.2 Design-time and effective signatures

Let $S[f, h] = 1$ when hypothesis h perturbs feature f. Write $\mathrm{closes}(h, g)$ when h drives gate g closed, and define

$$S_{\text{eff}}[f, h] = S[f, h] \wedge \neg\big(\exists\, g \in \mathrm{gate}(f) : \mathrm{closes}(h, g)\big). \tag{3}$$

A feature counts as evidence for h only if h perturbs it and does not silence it. Since $S_{\text{eff}} \le S$ entrywise, the induced partition $\Pi(S_{\text{eff}})$ is no finer than $\Pi(S)$. This is monotonicity of distinguishability under deletion of observations and we do not present it as more than that.

Applied to this system, with $\kappa = (H - P)/(H - 1)$ for $H = 8$ hypotheses and P observable classes: $\kappa(S) = 0.000$ over 8 classes, and $\kappa(S_{\text{eff}}) = 0.143$ over 7 classes, the merged pair being `gps_fault` and `nominal`. The design-time artifact reports full isolability for a system that does not have it.

Hypothesis	LOW feature	Gated by	Closed by this hypothesis?
`nominal`	(empty by definition)	—	—
`gps_fault`	`phys_gps`	`gvalid`, `gps_warmup`	yes, `gvalid`
`imu0_bias_drift`	`redundant_imu`	—	no
	`accel_motor_current`	`armed_gate`	no
	`attitude_gravity_mag`	—	no
`imu1_bias_drift`	`redundant_imu`, `accel_esc_rpm`	`armed_gate` (2nd)	no
`mag_interference`	4 features	`gvalid`, `speed_gate` (1 of 4)	no
`power_fault`	5 features	`armed_gate` (2 of 5)	no
`vibration_anomaly`	3 features	—	no
`unexpected_maneuver`	`temp_gps`	— (ungated)	no

Table 3: S and S_{eff} in one table. `gps_fault` has one feature, that feature is gated by `gvalid`, and a GPS fault closes `gvalid`. Its S_{eff} column is empty, which is `nominal`'s column.

3.3 Decision-relevant pairs

Following [**?**], only distinctions that change the commanded response matter. Reading the failsafe automaton's action table including action parameters, 27 of the 28 hypothesis pairs are decision-relevant. The single exception is `nominal` versus `unexpected_maneuver`: both command the empty action set. Note that `imu0_bias_drift` and `imu1_bias_drift` raise the same condition but parameterise `SELECT_IMU` differently, so that pair is decision-relevant.

4 The self-occlusion measurement

Three arms, identical flight profile and seed, differing only in whether the GPS fault destroys the fix. The failsafe is the shipping policy, driven from the batch trust history by the adapter the project already ships for its dashboard, so no policy logic is reimplemented.

	A: spoof, fix valid	B: denial, fix invalid	C: spoof then denial
`gvalid` closed, post-onset	0%	**62%**	50%
`gps_fault` on top	66.5%	**0.0%**	15.6%
`nominal` on top	33.5%	**100%**	84.4%
Time to first correct lead	1.44 s	**never**	1.44 s
Seconds `nominal` on top	13.4	**40.0**	33.8
with GPS trust ≥ 0.85	12.7	**40.0**	12.7
GPS trust minimum	0.5905	**0.9767**	0.7003
Maximum failsafe level	CAUTION	CAUTION	CAUTION
Failsafe actions issued	3	3	3

Table 4: Self-occlusion measured on the deployed pipeline. Arm A is the control: the spoof leaves the fix valid, the evidence survives, and the fault is named within 1.44 s. Arm B is the phenomenon.

Arm B. The fault is never named. For the full 40 s the diagnosis reports `nominal`, and the trust coefficient reads 0.9767 because the belief update is frozen by the freshness rule rather than driven down. Two independent channels report health simultaneously.

Arm C, the operationally interesting one. A spoof that first reveals itself and then escalates to denial is correctly diagnosed at 1.44 s and then returns to `nominal` for 84.4% of the remaining window. An adversary who is detected and then denies the signal restores the appearance of health.

What still works. Maximum failsafe level is CAUTION in all three arms with three actions issued in each. In arm B that escalation comes from the freshness disjunct, not the diagnosis.

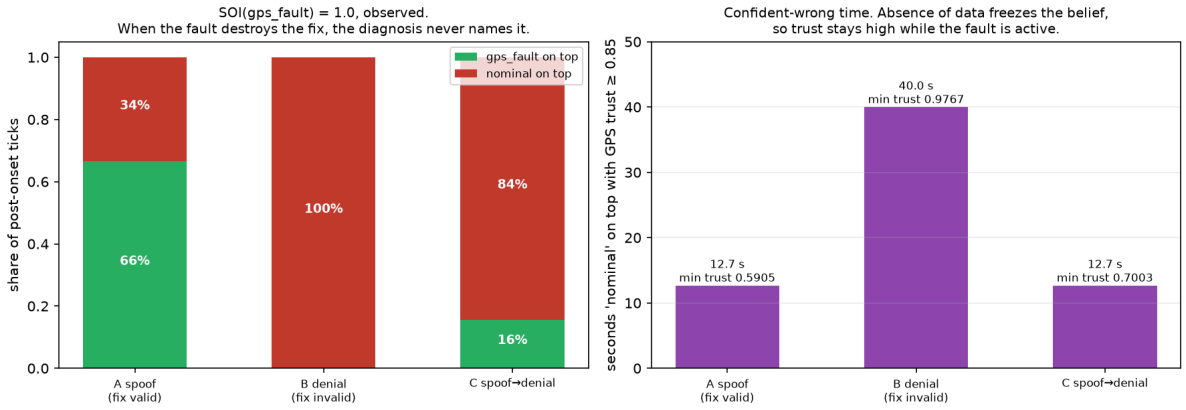


Figure 1: Left: share of post-onset ticks by leading hypothesis. Right: seconds spent reporting `nominal` with GPS trust above the actionable threshold, with the trust minimum annotated.

5 Containment: what is established and what is not

Two conditions must be kept apart. Write E_h for the features hypothesis h actually references in the deployed signature matrix, U_h for the set of all physically admissible relations that could bear on h, and C_h for the set of features controlled by gates that h itself closes.

$$E_h \subseteq C_h \qquad \text{present support silenced} \tag{4}$$
$$U_h \subseteq C_h \qquad \text{whole universe silenced} \tag{5}$$

Condition (4) gives effective undetectability now, for the system as built. Condition (5) would give irreparability: no addition of evidence could ever repair the collapse. **We establish (4) and we do not establish (5).** An earlier draft of this work did not separate them.

(4) **holds for `gps_fault`, by inspection.** $E_{\texttt{gps_fault}} = \{\texttt{phys_gps}\}$, read from the signature matrix in `bayes.py`. `phys_gps` is gated by `gvalid` $\cdot \mathbb{1}[\tau > 3]$, read from `online.py`, so `phys_gps` $\in C_{\texttt{gps_fault}}$. A one-element subset check is the whole proof. This is what Section 4 measures the consequence of.

(5) **is not established, and we drop it permanently.** U_h has no absolute definition. It is definable only relative to a triple (M, S, R): a measurement set M the platform actually provides, a signal model S that says what those measurements mean, and a relation language R that says what may be built from them. Change any of the three and U_h changes. There is therefore no unrestricted irreparability claim to be made, and we withdraw the one an earlier draft of this work made. What remains is containment relative to a stated (M, S, R), which is what Section 5.3 reports.

Containment is cheap to check, and we do not claim otherwise. Deciding $E_h \subseteq C_h$ is a subset test over two finite sets, linear in the size of the signature matrix and the gate map. The design-rule audit of Section 11 runs it over all 19 signature entries in the project's test suite on every commit. Nothing here is computationally hard, and any framing that suggests otherwise would be wrong.

5.1 A generator-dependent sensitivity study

A natural objection is that arm B is an artifact of a sparse signature matrix: this system averages 2.4 features per hypothesis, and `gps_fault` has exactly one. If the collapse disappears once hypotheses carry more evidence, the finding is about sparsity and not about gating.

We tested this by synthesising signature matrices over the same gate structure at densities from 1 to 8 features per hypothesis, 400 draws per cell, with one designated hypothesis closing `gvalid`. Two arms differ only in where that hypothesis draws its features from.

Features per hypothesis	Arm A: uniform draw		Arm B: draws from gated set	
	mean gap	$P(\text{gap} > 0)$	mean gap	$P(\text{gap} > 0)$
1	0.0264	0.185	0.1018	0.713
2	0.0050	0.037	0.1400	0.980
3	0.0011	0.007	0.1425	0.998
4	0.0004	0.003	0.1425	0.998
5	**0.0000**	**0.000**	**0.1429**	**1.000**
6–8	0.0000	0.000	**0.1429**	**1.000**

Table 5: The gap $\kappa(S_{\text{eff}}) - \kappa(S)$ against signature density. Arm A collapses to zero by density 5. Arm B saturates at $1/7$, this system's value, at every density.

Arm A confirms the objection for a generator that draws hypothesis evidence freely: the gap vanishes with density. Arm B does not: it saturates at $0.1429 = 1/7$ with the designated hypothesis fully silenced in 100% of draws at every density.

What this does and does not show. It is a statement about two generators, not about the world. Arm A shows that the feature-density argument against arm B succeeds when the generator draws freely. Arm B shows that the same argument fails when the generator enforces containment. Which generator describes a real system is a separate question, and the deployed system's is arm B's for `gps_fault` by (4). It is a sensitivity study, and it is not evidence for (5).

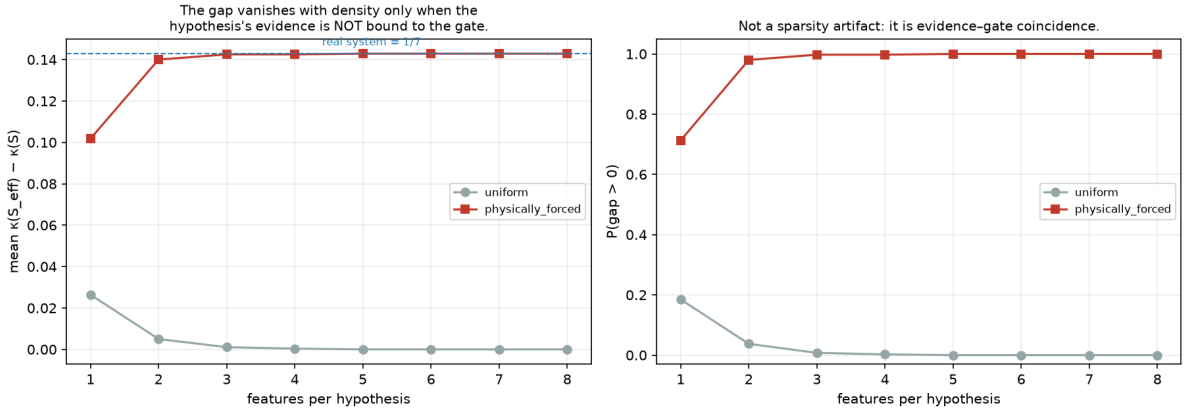


Figure 2: The density sweep. Left: mean gap. Right: probability the gap is positive.

5.2 The interior is a threshold, not a slope

Arms A and B are corners. We swept the interior: let a fraction ρ of the designated hypothesis's features lie inside C_h and $1 - \rho$ outside, at densities 1, 2, 3, 4, 6 and 8, 400 draws per cell.

The result is a threshold at $\rho = 1$. With every feature inside C_h the hypothesis is absorbed into the nominal class in 100% of draws; with *exactly one* feature outside, it is absorbed in 0% of draws. The drop is 1.000 at every density tested. Effective undetectability does not degrade gracefully with partial containment; it requires full containment of the present support.

The engineering statement this licenses is narrower and more useful than anything about universes: **one ungated relation suffices**. That is a checkable design requirement, and Section 12 costs it out.

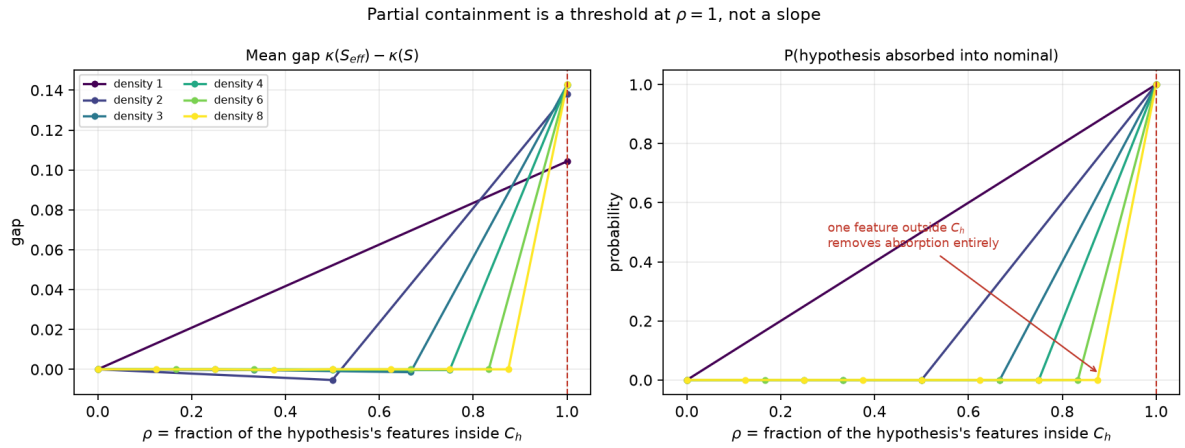


Figure 3: The interior of the containment sweep. Left: mean gap against ρ. Right: probability the designated hypothesis is absorbed into the nominal class. The transition at $\rho = 1$ is a step at every density.

A correction to our own reasoning. An earlier note in this project justified $\Pi(S_{\text{eff}})$ being no finer than $\Pi(S)$ by observing that $S_{\text{eff}} \leq S$ entrywise, so equal columns stay equal. That justification is wrong in general. Entrywise domination preserves column equality under *uniform* row deletion, but per-hypothesis silencing is *column-dependent*: two hypotheses sharing a column in S separate in S_{eff} when only one of them silences, and the partition gets finer. Searching 200 000 random draws found 483 counterexamples, a rate of 0.24%. The deployed matrix is unaffected, because `gps_fault` is the only hypothesis that silences anything and its column is unique, so no shared column can be broken. The computational verification on the real matrix stands; the general argument does not.

5.3 A bound on the universe over the ingested field set

We cannot enumerate $U_{\texttt{gps_fault}}$. We can enumerate the relations derivable from the telemetry fields the ingestion path actually reads, because that field set is finite and closed. Reading the field list mechanically from `ulog_source.py`, six relations in the edge library span the GPS node.

Formal containment over this field set is therefore *false*: one relation sits outside. But a relation outside C_h is only an escape if its value responds to the fault. Running the deployed pipeline clean and under denial on the same truth and seed, `temp_gps` consistency over the fault window is bit-identical between the two runs (mean 0.9155, min 0.640319, max 1.0 in both) while the GPS channel's fresh fraction falls from 1.0 to 0.001597. The temporal predictor is computed from the channel history and never consults channel validity, so a dropout leaves the underlying

Relation	Spans	In $C_{\texttt{gps_fault}}$
`altitude_baro`	baro, gps	yes
`velocity_gps_doppler_0`	imu0, gps	yes
`velocity_gps_doppler_1`	imu1, gps	yes
`heading_course`	mag, gps	yes
`phys_gps`	gps	yes
`temp_gps`	gps	**no**

Table 6: GPS-spanning relations over the currently ingested field set. Five of six are `gvalid`-gated.

series and therefore the residual untouched. `temp_gps` is ungated and carries no information about denial.

The scoped statement we can defend: over the currently ingested telemetry field set, formal containment is incomplete by one relation and *effective* containment is complete. This is weaker than (5) and much stronger than nothing, and it is defensible because the enumeration is mechanical and the field set is closed.

It also names the escape route precisely. Every GPS-spanning relation buildable from the ingested fields depends on position, velocity or altitude, all of which are meaningless without a fix and are gvalid-gated by construction. The fields that could support a relation surviving fix loss are `satellites_used`, `noise_per_ms`, `jamming_indicator`, `hdop`, `vdop`, `s_variance_m_s` and `eph`/`epv`, and the ingestion path reads none of them.

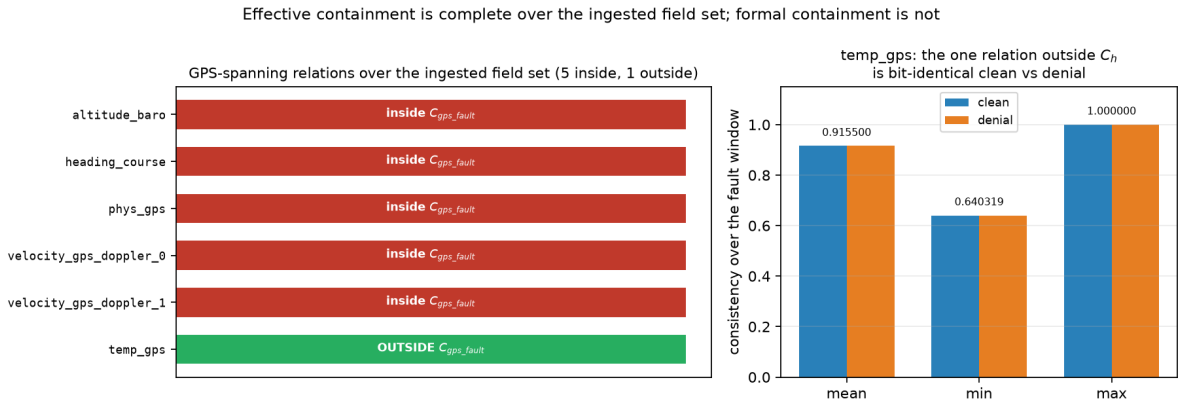


Figure 4: Left: the six GPS-spanning relations, by whether `gvalid` controls them. Right: `temp_gps`, the one relation outside $C_{\texttt{gps_fault}}$, takes bit-identical values clean and under denial.

6 Gate closure frequency, measured against a real autopilot

Everything above is structural: it says what can happen, not how often. The quantity that converts it into a risk statement is $q_{\text{gvalid}} \mid h = P(\texttt{gvalid}\text{ closed at a tick} \mid h)$, and it cannot be estimated from the public log corpus because that corpus is unannotated and the conditioning variable does not exist in it.

We obtained the label by controlled injection instead. A full ArduCopter autopilot runs in software-in-the-loop with its own EKF, its own GPS driver and its own MAVLink stream; the deployed trust layer ingests that stream through the production adapter, and the gate trace is the layer's own `gvalid` read out of its per-tick edge record rather than recomputed. Four arms of six flights each share the same six flight profiles, so the design is matched. Each flight arms, climbs to 20 m, flies a square, and receives its injection at $t = 60$ s; the conditioning window is

[60, 85] s. Pre-registration, stopping rules and the estimator follow a specification fixed before the campaign ran.

Arm	Monitored ticks	Closed	q_{gvalid}	Injection reached the receiver
`nominal`	3744	0	**0.000**	(no injection)
`denial`	3744	3684	**0.984**	satellites 9.93 → 3.05
`glitch`	3744	0	**0.000**	position step 1.1 → 62–76 m
`link loss`	3744	3672	**0.981**	satellites 9.93 → 0.17

Table 7: Gate closure by fault class, 6 flights per arm. The right column is the control that distinguishes a null result from an injection the autopilot ignored.

A correction to our own arm labels. The fourth arm was originally called `degraded` and described as partial constellation loss, because it combined a reduced satellite count with receiver byte loss. A follow-up sweep (Section 6.2) shows the satellite count contributes nothing: the gate does not close at any satellite count. The closure came entirely from the byte loss, which is a receiver link failure, so the arm is relabelled. The number is unchanged; the attribution was wrong.

Three things follow, and the third is the one that changes the paper.

The gate never closed in healthy flight. Zero closures in 3744 monitored ticks. By the pre-registered stopping rule a zero denominator makes the ratio unbounded above, so we report the one-sided statement only and no interval. This is also the weakest of the three results: the simulated receiver does not spontaneously lose a fix, so a nominal rate of zero is a property of the generator before it is a statement about flight.

Denial closes the gate, end to end, on a real autopilot. $q_{\text{gvalid}} \mid \texttt{denial} = 0.984$, and the diagnosis over the same window reports `nominal` at $p = 0.977$ with `gps_fault` at $p = 0.0227$, while the failsafe reaches CAUTION at $t = 61.5$ s through the freshness disjunct. That reproduces Section 4 on a different generator, with a different vehicle model, a different EKF and the production ingestion adapter in the path. It is confirmation of the mechanism, not evidence about frequency: `SIM_GPS_DISABLE` switches the fix flag off, so a high closure rate under it is close to tautological.

A GNSS fault that keeps the fix does not close the gate. This is the informative arm. A position glitch of 62 to 76 m, with ten satellites still reported and the fix retained, produced **zero** gate closures in 3744 ticks. For that fault class the design-time and effective partitions coincide, so the self-occlusion of Section 4 does not arise at all: the evidence stays available.

The layer did not use it. GPS trust fell only to 0.906–0.934 against a `nav_caution` threshold of 0.75, no failsafe transition occurred, and the diagnosis stayed at `nominal` $p = 0.9997$. **That is a different failure mode from the one this paper is about** and we do not merge them: the first is evidence destroyed by gating, the second is evidence present and not acted on. Section 7 tests its cause.

What this bounds. Self-occlusion is specific to fix-loss faults, not to GNSS faults in general. Both arms that removed the fix closed the gate; the arm that kept it did not. The scope of Section 4 is therefore narrower than an unqualified reading would suggest, and this campaign is what narrows it.

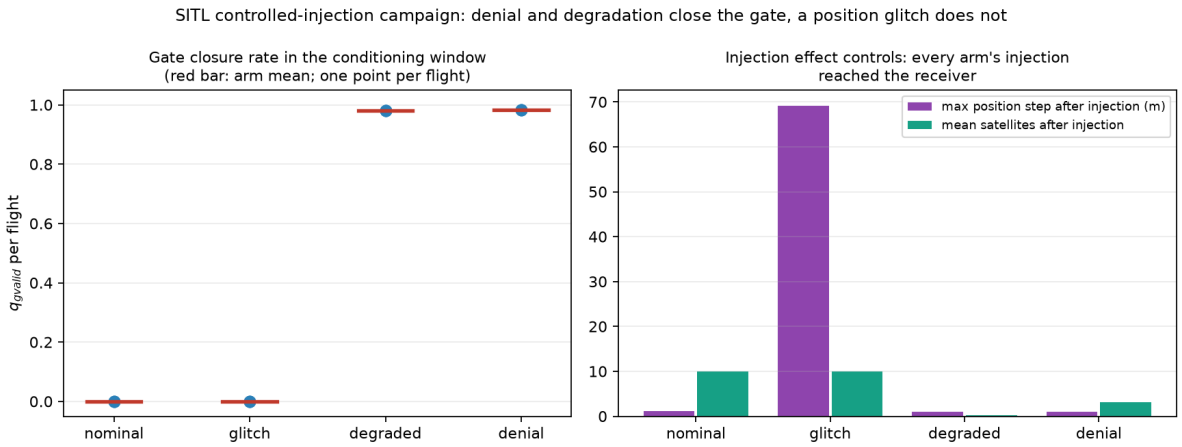


Figure 5: Left: per-flight closure rate by arm. Right: the injection-effect controls.

Limits of this campaign, stated with the result. The per-flight statistic is identical across flights in every arm, so the bootstrap interval collapses to a point. That interval is a statement about simulator determinism and not about estimate precision, and we report it as degenerate rather than as narrow. The receiver is simulated throughout, and nothing here is a measurement of real GNSS environments.

6.1 What varied, checked rather than assumed

Zero between-flight variance is not a normal experimental result, so before building on the campaign we tested whether the varied parameters reached the simulation at all. If they had not, the campaign would be one condition replicated 24 times and every number above would be wrong.

- **Home site: real.** Six sites produced mean attitude differences with a between-site standard deviation of 0.0127 rad against a within-site standard deviation of 5.9×10^{-5} rad, measured by flying one site three times. The ratio is 216, so the sites are genuinely distinct conditions.
- **Wind: cosmetic, and this corrects our own description.** `SIM_WIND_SPD` is accepted and reads back correctly at every value, but a vehicle commanded to hold position tilts 0.159° at 0 m/s and 0.159° at 12 m/s. No wind force is applied in this build. The campaign was described as varying two axes and varied one.
- **Gate response: deterministic.** First closure occurs 0.44 s after injection in every denial flight. The injection dominates the gate regardless of where the vehicle is, which is why the between-flight variance is zero. It is a property of the gate response, not an artifact of the experiment.

The marginal value of additional flights follows from this directly: bootstrap interval width is zero at every n from 2 to 12, because there is no between-flight variance to resample. Flying more of the same condition buys nothing. The binding constraint is condition diversity, which is what the rest of this section spends its flights on.

6.2 The satellite ladder, and a fault the layer cannot see at all

`SIM_GPS_NUMSATS` was swept from 9 down to 0, one flight per rung. **The gate never closed.** At every rung, including zero satellites, the receiver continued to report a 3D fix on all 578 samples and $q_{\text{gvalid}} = 0.000$, while `satellites_visible` tracked the commanded value exactly.

Two readings, and both belong in the record.

The first is a fidelity limit: a real receiver does not assert a 3D fix on zero satellites, so this simulator's fix flag is not coupled to its own satellite count, and the ladder does not tell us where a real receiver would flip.

The second is the paper's point demonstrated end to end. The injected fault was fully present in the telemetry, in a field the receiver was reporting correctly, and the layer never saw it: `satellites_used` is not among the fields the ingestion path reads, and the gate is keyed on fix validity alone. The diagnosis reported `nominal` throughout. Nothing was unobservable; the observation was not routed.

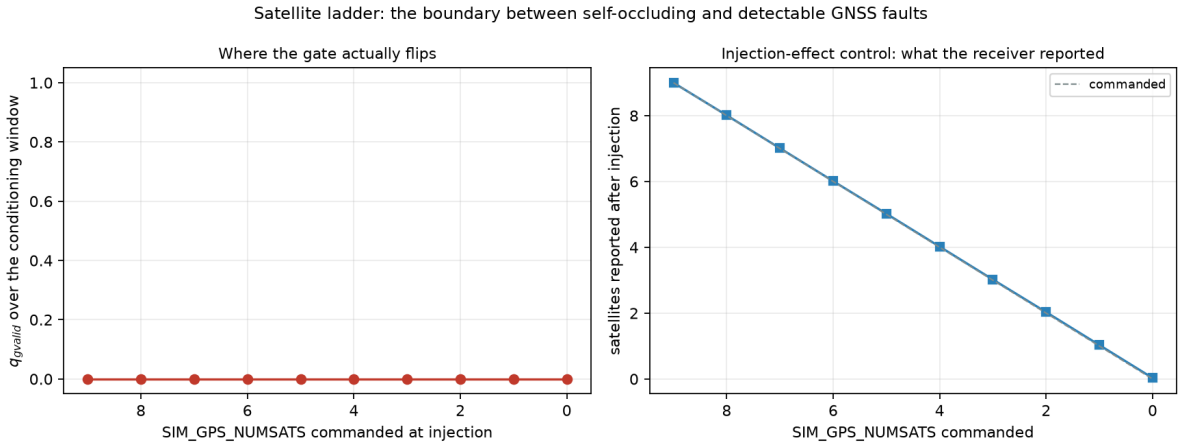


Figure 6: Left: gate closure against commanded satellite count; the gate never closes. Right: the injection-effect control, showing the receiver reported the commanded count faithfully at every rung.

6.3 Replication along onset and duration

Denial injected during climb, early cruise, mid cruise and late cruise gives q_{gvalid} of 0.984, 0.984, 0.984 and 0.982, with first closure 0.44 s after onset in all four. The response does not depend on mission phase.

Denial lasting 2, 5 and 15 s and then restored while airborne gives q_{gvalid} of 0.088, 0.208 and 0.609 over the fixed 25 s window, which is the denial duration divided by the window in each case. Closure latency is 0.44 s and reopen latency 0.64 s throughout. Gate occupancy tracks denial duration exactly, with sub-second latency at both edges.

7 The glitch non-alarm has a tested cause

The `glitch` arm leaves the fix valid, so the gate stays open and the evidence stays available. The layer did not escalate anyway. We tested why.

The hypothesis we started from, and its fate. The natural explanation is that the EKF absorbed the step into its state, the innovation stayed small, and a residual monitor saw nothing. That is why RAIM operates in the measurement domain rather than the position domain. It splits into two claims and they do not have the same answer. The EKF *did* absorb the step: its fused position ends up following the glitched GNSS position by a fraction of 0.79 to 0.998 across the sweep. Its innovation did *not* stay small: reported horizontal position variance rises from about 0.006 to between 1.2 and 1113, scaling with glitch size. And neither half would have

explained the non-alarm, because this layer reads the raw receiver output and never consumes the fused estimate.

What the sweep shows instead. Step glitches from 5.6 m to 5.6 km:

Position error (m)	Min GPS trust	Min physics residual	Ticks at floor	Alarm
5.6	0.9237	0.0000	26	no
22.3	0.9089	0.0000	32	no
55.7	0.9063	0.0000	33	no
222.6	0.9061	0.0000	33	no
1113.2	0.9061	0.0000	33	no
5566.0	0.8881	0.0000	40	**no**

Table 8: The residual detector floors at every magnitude, including the smallest. No step magnitude produces an alarm.

The detector is not blind. It saturates at zero for every injected error, including 5.6 m, so the layer sees every glitch. What bounds the trust excursion is not magnitude but *duration*: the step detector decays at 0.85 per tick, so a step floors the residual for 26 to 40 ticks whatever its size, and the trust recursion integrates over a window long enough that this never crosses the threshold. Trust saturates near 0.906 from 55 m upward.

There is no step magnitude at which this system alarms. A GNSS position error of five kilometres, arriving as a step, is seen by the detector and never escalated.

And a ramp inverts the ordering. The same total 222.6 m offset applied as a 5, 20 or 60 s ramp does alarm, with minimum trust 0.717 to 0.723 and 104 to 168 ticks at the floor. A slow error holds the residual above threshold long enough to move the recursion; a fast one of the same size does not. This is the opposite of what one would expect from an estimator-tracking argument, and it is the opposite of the EKF's own behaviour: the 20 and 60 s ramps are absorbed completely, with EKF variance staying near baseline. Each monitor is blind to what the other sees.

7.1 Why, in closed form

The trust coefficient is a Beta belief with adaptive forgetting (`engine.py`). Two facts fix the behaviour.

First, the belief mass $S_k = \alpha_k + \beta_k$ obeys $S_{k+1} = \rho S_k + 1$ regardless of the evidence, so it converges to $1/(1-\rho)$. Calm flight has zero spread between evidence components and runs at $\rho_{\max} = 0.998$, so the belief settles carrying $S_0 = 1/(1-\rho_{\max}) = 500$ units of accumulated agreement.

Second, when a fault destroys one evidence component the spread jumps to 1 and ρ drops to its floor $\rho_{\min} = 0.95$, whose sustainable mass is only $M = 20$. Holding evidence at a constant e_f for N ticks then gives, with $x = \rho^N$,

$$T_N = \frac{xS_0T_0 + (1-x)Me_f}{xS_0 + (1-x)M} \tag{6}$$

a mass-weighted average of the old trust and the new evidence. Two corollaries follow, and both are measured above.

The evidence floor. At the GPS node the fusion weights are $w_{nbr}, w_{phys}, w_{temp}, w_{stat} = 0.35, 0.30, 0.20, 0.15$. A fault that destroys only the physics residual leaves $e_f = 0.35 + 0.20 +$

$0.15 = 0.70$, against a threshold of 0.75. Trust cannot fall below 0.70 however large the error. Measured minimum fused evidence in the saturated cells of the sweep: **0.7000**.

The ticks required. Setting $T_N = T^*$ in (6),

$$N^* = \frac{\ln\left(\frac{M(T^*-e_f)}{S_0(T_0-T^*)+M(T^*-e_f)}\right)}{\ln \rho} \approx 93 \text{ ticks} = 3.7\,\text{s} \tag{7}$$

Note which term dominates: $S_0(T_0 - T^*) = 125$ against $M(T^* - e_f) = 1.0$. **The delay is set by how much agreement the calm period banked, not by how bad the fault is.** Halving the concentration cap nearly halves the delay; doubling the threshold margin barely moves it.

Escalation additionally requires the failsafe's 1 s dwell, so the boundary is $N^* + 25 \approx 118$ ticks at the floor. Against the 21 cells of the sweep this classifies **19 correctly**. Both misses lie near the line, at $+0.7$ and -12.3 ticks, and the model assumes a constant e_f where the measured floor ranges 0.678 to 0.700.

Equation (6) was checked against the shipping recursion driven with a synthetic evidence sequence: maximum absolute error **0.00** over 200 ticks, and the predicted crossing at 93.3 ticks against an observed crossing at tick 94.

And why magnitude cannot substitute for duration. The step detector holds a peak decaying at 0.85 per tick, so the residual stays saturated for $N_{floor} \propto \ln(\text{amplitude})$ ticks. A thousandfold larger error buys at most 42 extra ticks, every one of them on a residual already pinned at zero and therefore contributing no further evidence. **A saturating detector with a fixed decay cannot escalate on magnitude alone.**

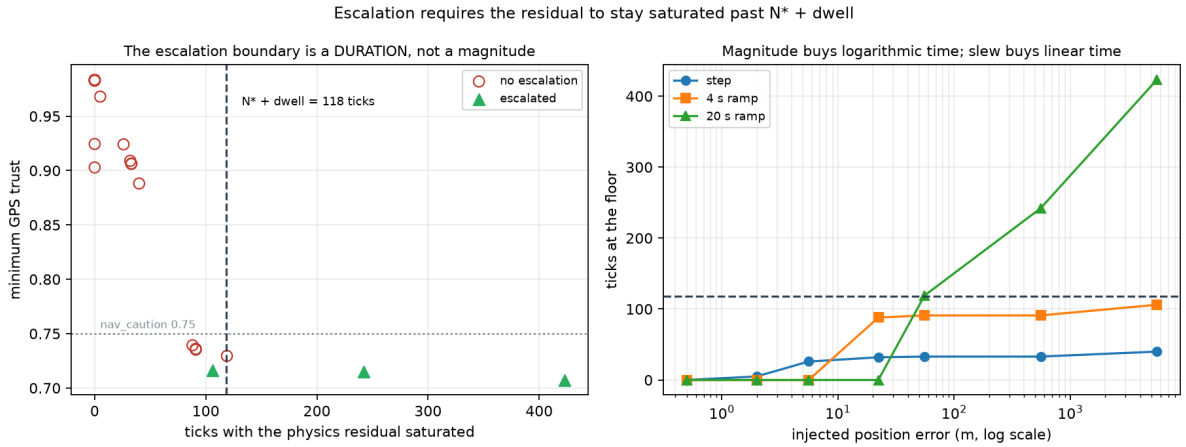


Figure 7: Left: escalation against time at the floor, with the predicted boundary. Right: how each axis buys time.

8 Two detectors, and what neither sees

The vehicle carries two independent monitors of GNSS integrity: its own estimator, which gates measurements on innovation test ratios, and the diagnostic layer, which compares raw receiver output against inertial and barometric predictions. They read different signals, so their blind spots need not coincide. We swept amplitude against slew rate, 21 cells, recording for each whether the estimator flags, whether the layer's residual saturates, and whether the layer escalates.

Coverage	By detection	By escalation
both	12 / 21	3 / 21
estimator only	0 / 21	9 / 21
layer only	1 / 21	0 / 21
neither	**8 / 21 (38%)**	**9 / 21 (43%)**

Table 9: Two-detector coverage over the amplitude-slew grid.

The uncovered region is not empty and it is not a corner. It is a wedge that widens as the error slows: an instantaneous step is caught above 2 m, a 4 s ramp is missed up to 5.6 m, and a 20 s ramp is missed up to 22.3 m. The estimator misses slow errors because they are absorbed into its state with small innovations; the layer misses them because its step detector never saturates. Both blind spots move the same way, so they overlap instead of covering each other.

Judged by escalation rather than detection the picture is worse: the layer escalates in 3 of 21 cells and the uncovered fraction rises to 43%. A detection nobody acts on is not coverage.

We state the limits of this plainly. A 2 m error is inside normal GNSS noise and not detecting it is correct behaviour, not a blind spot. The result is the *shape* of the boundary rather than any single cell: the uncovered amplitude grows with slew time, and nothing in the grid closes it. The largest uncovered error we measured is 22.3 m over 20 s, which is a regime a gradual position walk-off would occupy. We measured one autopilot build and one diagnostic layer, and we make no claim about detectors in general.

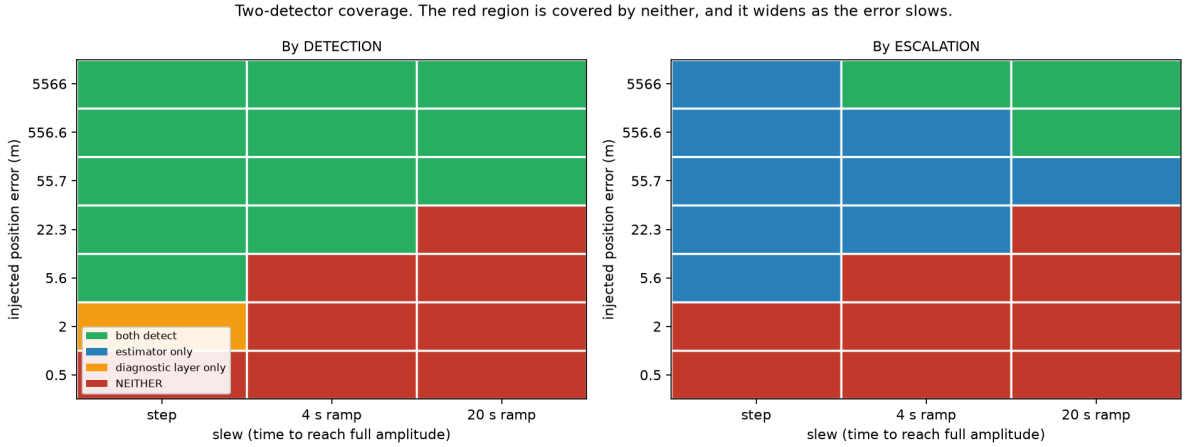


Figure 8: Two-detector coverage by detection and by escalation.

Flight phase. Injecting the same 556.6 m 4 s ramp during climb, cruise and descent gives minimum trust 0.697, 0.735 and 0.744; only the climb case escalated. The dependence is real but small, and it is smaller than the slew dependence.

9 False-healthy exposure, and the cost of abstaining

The quantity that matters operationally is not whether a gate closes but how long the system asserts health while holding nothing it is entitled to assert it from, and for how much of that time it is actually wrong. Those are two different questions and Section 9.2 separates them into four.

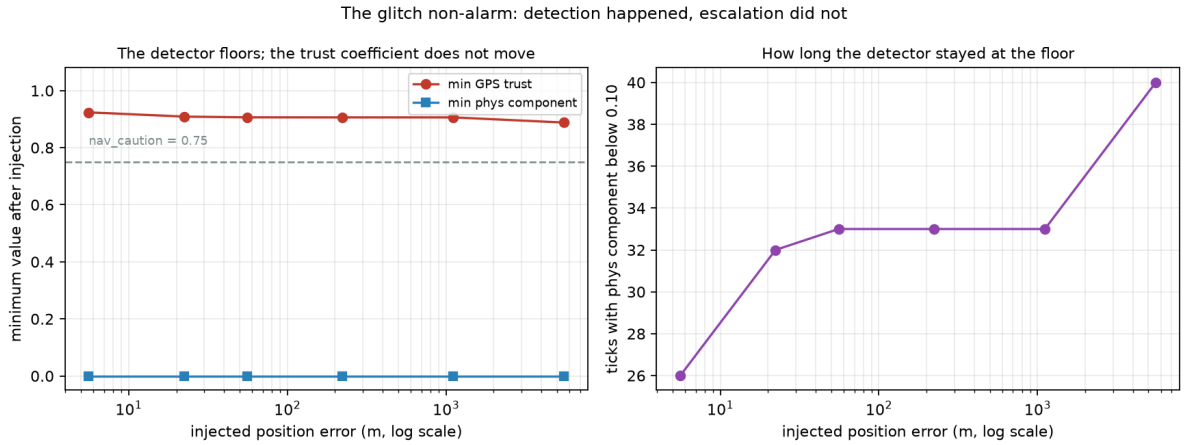


Figure 9: Left: the residual detector floors at every magnitude while the trust coefficient stays far above threshold. Right: how long the floor lasts, which is what bounds the excursion.

9.1 The four states, and what this system emits instead

The implementation has no `NOT_APPLICABLE` state. Reading what it emits in each case, from source, is the finding.

State	Condition	What is emitted
`HEALTHY`	failsafe NORMAL and top cause `nominal`	emitted
`FAULT`	failsafe above NORMAL, or top cause not `nominal`	emitted
`UNKNOWN`	no evidence available, Beta belief held	**emitted as** `HEALTHY`
`NOT_APPLICABLE`	hypothesis support entirely silenced	**emitted as** `HEALTHY`

Table 10: The last two rows are the gap. Both states that mean "I cannot assess this" surface as health.

Two lines of source produce the last two rows. In the diagnosis, a gated feature is skipped outright, so when a hypothesis's entire support is gated its likelihood ties with `nominal` over the surviving features and the prior decides; the normalized hypothesis weight concentrates on health. In the trust recursion, a node that loses every evidence source has its Beta belief held rather than driven down, which is the right choice given the available vocabulary. A held belief and a confirmed one are the same number at the output.

This is a semantics gap, not an observability limit. The evidence for GNSS denial exists at the receiver and is well studied; this system routes its diagnosis through position-domain relations that the fault switches off, and then has no way to say so.

9.2 Four quantities, because one was not enough

A single exposure number conflated two different things. Time spent healthy with no applicable evidence is an *unsupported* verdict; without ground truth it may still be correct by accident, and calling it false overclaims. Controlled injection supplies the ground truth an operational log cannot, so the quantity splits. Let $A_h = 1$ mean the test for h is applicable at this tick and $F_h = 1$ mean the injection schedule says h is truly present:

$$
\begin{aligned}
T_{NA} &= \text{time}\{A_h = 0\} && \text{inapplicable} \\
T_{UH} &= \text{time}\{A_h = 0 \wedge \text{healthy}\} && \text{unsupported healthy} \\
T_{FH} &= \text{time}\{F_h = 1 \wedge \text{healthy}\} && \text{false healthy} \\
T_{SOFH} &= \text{time}\{F_h = 1 \wedge A_h = 0 \wedge \text{healthy}\} && \text{self-occluded false healthy}
\end{aligned}
$$

Only T_{SOFH} is this paper's original subject. Verdicts are reported in the ISO/IEC 9646-1 conformance vocabulary, PASS, FAIL, INCONCLUSIVE and INAPPLICABLE, presented as HEALTHY, FAULT, UNKNOWN and NOT_APPLICABLE. That unusable data must not be presented as normal data is established at field level: ARINC 429 carries a Sign/Status Matrix distinguishing Failure Warning and No Computed Data from Normal Operation, and OPC UA carries a StatusCode whose Bad severity means the value must not be used. We are aware of no standard term for the exposure quantity itself and do not coin one.

9.3 Measured exposure

Four arms plus a recovery arm, three flights each, 25 s conditioning window, attribution channel unless stated. All policies are scored from the same tick stream, so the comparisons are paired.

Arm	F_h	T_{NA}	T_{UH}	T_{FH}	T_{SOFH}	T_{FH} status	T_{FH} abstain
`nominal`	no	0.00	0.00	0.00	0.00	0.00	0.00
`glitch`	yes	**0.00**	**0.00**	**24.64**	**0.00**	24.64	**24.64**
`link loss`	yes	24.48	24.48	24.96	**24.48**	1.52	0.48
`denial`	yes	24.56	23.97	24.37	**23.97**	1.44	0.40
`denial, recovered`	yes	15.20	14.61	14.37	13.97	1.44	0.40

Table 11: Seconds in a 25 s window. The `glitch` row is the point of the decomposition.

The `glitch` row is why one number was not enough. It has the *largest* false-healthy exposure of any arm, 24.64 of 25 s, and $T_{SOFH} = 0$ exactly. $A_h = 1$ for every tick: the fix is retained, `gvalid` stays open, `phys_gps` keeps contributing. The hypothesis is applicable throughout, so no part of that exposure can be self-occlusion. Its cause is detector sensitivity and the escalation interface, which Section 7 characterises separately. Pooling it with the denial arm under one heading would have attributed the programme's largest exposure to the wrong mechanism.

Two channels, because reporting either alone would mislead. On the attribution channel the denial arm spends 23.97 of 25 s asserting `nominal` with no applicable evidence. On the reported status it spends 1.44 s, because the freshness disjunct drives the failsafe out of NORMAL without consulting the diagnosis. The escalation path holds. The naming path does not.

A permanent limit on operational data. F_h does not exist in unannotated telemetry. T_{NA} and T_{UH} are estimable from the 26-log corpus; T_{FH} and T_{SOFH} never will be. That is a property of the data, not of the method, and it is why controlled injection is not optional here.

9.4 The abstaining baseline

We implemented, *in research code only and with no production change*, a counterfactual policy that emits `NOT_APPLICABLE` when a hypothesis's entire support is silenced. Scored on the same flights:

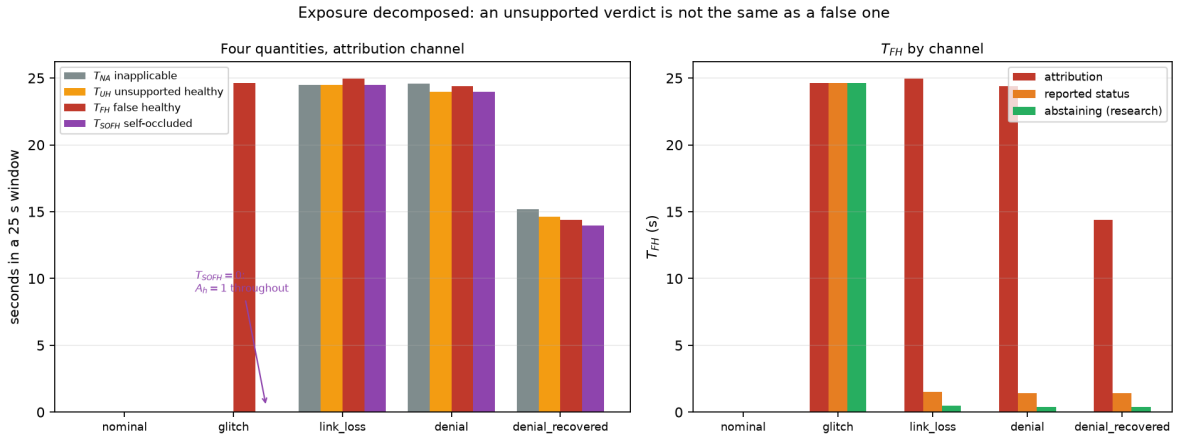


Figure 10: Left: the four quantities by arm on the attribution channel. Right: T_{FH} by channel, showing that the abstaining policy removes the denial and link-loss exposure and does nothing at all for the glitch.

- **It removes the self-occlusion exposure completely.** T_{SOFH} falls to 0.00 s in every arm, and T_{FH} falls from 24.37 to 0.40 s under denial and from 24.96 to 0.48 s under link loss.
- **It does nothing for the glitch.** T_{FH} stays at 24.64 s, unchanged to two decimals. There is nothing to abstain about when $A_h = 1$ throughout. *The two defects need two different fixes, and this is the measurement that shows it.*
- Onset of a non-healthy status moves from 1.48 s to **0.44 s** under denial and 1.56 s to 0.52 s under link loss. The abstention is available as soon as the gate closes, without waiting for the freshness watchdog.
- **Nuisance cost is zero.** In `nominal` the policy never fires: zero silenced ticks, zero abstentions, behaviour identical to the deployed system.
- **Recovery costs nothing.** After denial ends while airborne, both policies return to health after 5.71 s, the failsafe's own de-escalation dwell.

On this evidence the abstaining state is one-sided for the faults it addresses, and irrelevant to the faults it does not. We report it as a measured counterfactual and not as a recommendation to ship, because three flights per arm against one simulator is not the basis for a production change.

10 Chatter resonance

Exponential smoothing with coefficient α_s has a characteristic time constant $1/\alpha_s$. A gate that toggles on that timescale should maximise label instability: faster toggling is averaged out, slower toggling lets the accumulator settle in each state. With $\alpha_s = 0.06$ the predicted peak is at $1/\alpha_s = 16.67$ frames.

We swept gate closure duration from 1 to 100 frames against the deployed diagnosis kernel, holding the evidence pattern fixed so that every label change is caused by availability alone. The observed peak is at 21 frames with 95 top-label changes per 2000 frames, against 1 to 3 changes at durations below 18. The prediction holds on the implementation.

A methodological note. Our first attempt at this sweep made every feature anomalous, which allowed a multi-feature hypothesis to lead whether the gate was open or closed, and produced

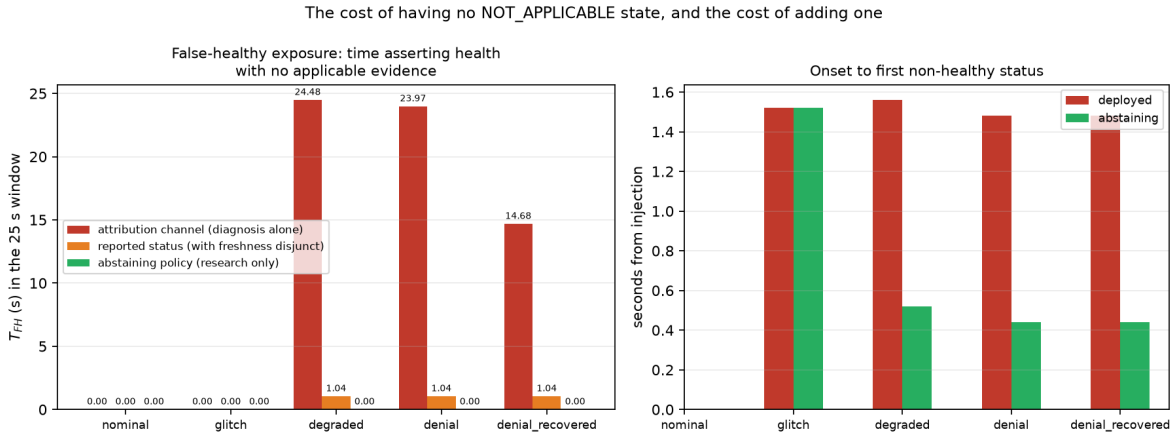


Figure 11: Left: false-healthy exposure by channel and policy. Right: onset to first non-healthy status.

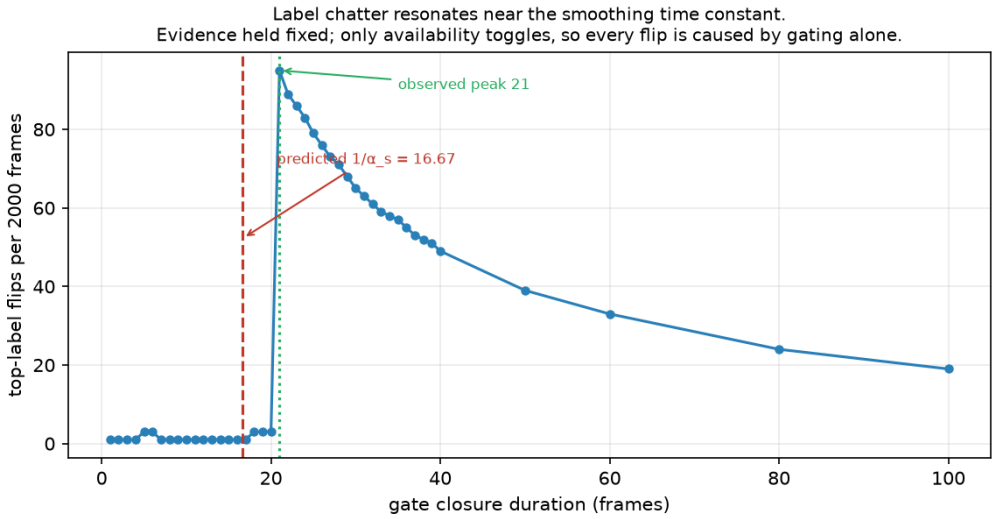


Figure 12: Top-label changes against gate closure duration. Evidence held fixed, so every change is caused by gating.

zero label changes at every duration. That null result was a property of the probe, not of the system. The gated feature must be decisive for the leading label or there is nothing to observe.

11 A design rule and its enforcement

One ungated relation suffices. That is the transferable claim, and it is the threshold result of Section 5.2 restated as engineering. Absorption of the hypothesis into the nominal class runs at 1.000 when every one of its features lies inside C_h and at 0.000 when exactly one lies outside, at every signature density from 1 to 8. There is no dose-response curve to tune against and no partial credit: the requirement is a single relation whose gate is independent of the fault, and the requirement is met or it is not.

That sharpness is what makes the rule checkable rather than advisory:

> A feature intended to discriminate hypothesis h must not be gated by a gate that h closes.

Auditing this system's 19 signature entries gives 18 clean and one violation: `gps_fault` / `phys_gps` / `gvalid`. The violation is fatal rather than merely weakening because `phys_gps` is that hypothesis's sole feature, so the effective signature empties rather than thins.

The rule is enforced in the project's test suite. The test pins the violation list exactly: a new violation fails the build, and the known violation disappearing also fails, so a silent repair cannot land without the record being updated in the same commit. This is the transferable engineering output of the work. It requires only a signature matrix and a gate-to-feature map, both of which any gated diagnostic system already has.

12 A repair, measured

A second collision existed: `imu0_bias_drift` and `imu1_bias_drift` shared the signature {`redundant_imu`} whenever the motor edges gated off at disarm, because their distinguishing features were both `armed_gate`-controlled. The repair is one entry: `attitude_gravity_mag` added to `imu0_bias_drift`. It compares magnetometer heading against `imu0`-integrated yaw, so an `imu0` bias corrupts it and an `imu1` bias does not, and it is ungated, which is the property that matters.

Measured before and after over the full configuration lattice:

	Before	After
$\kappa_{\max}$ at full instrumentation	0.2857	**0.1429**
Decision-relevant pairs exposed at full instrumentation	2	**1**
New collisions across 1232 reachable configurations	—	**none**
Pairs with an order-one cut set	7	**6**
Components in any order-one cut set	9	**7**
`imu0`/`imu1` minimum cut set size	1	**2**
Peak hypothesis weight for `imu0_bias_drift`	0.9861	**0.9203**

Table 12: The repair. Cut-set analysis follows [**?**, **?**].

The cost is real and we report it. Peak normalized hypothesis weight for the target hypothesis falls by 0.066, because a larger LOW set requires more features to be simultaneously anomalous. Detection latency and minimum trust are unchanged on all five scripted fault scenarios, the clean scenario produces zero false positives before and after, and the committed campaign artifact regenerates bit-identical.

We also checked the obvious concern that routing IMU discrimination through the magnetometer creates a new single point of diagnostic failure. It does not: `mag` was already an order-one cut set for one pair before the change and remains exactly that, while `armed_gate` and `motor_esc` dropped out of the order-one set entirely.

13 Limitations

The corpus is unannotated. The 26 public flight logs used elsewhere in this project carry no ground-truth health labels. No detection performance on real flight data is claimed anywhere in this paper. The measurements in Sections 4 and 10 are on simulated fault injection into the deployed pipeline.

Closure attribution is derived, not observed. closes(h, g) comes from reading the gate predicate in source plus the physical fault model. If a fault closes a gate in flight that the model says it does not, Equation 3 is wrong in the optimistic direction and the real gap is larger than reported.

Structural, not statistical. κ bounds structural indistinguishability. Two hypotheses in one observable class may still separate on residual magnitude or variance. In arm B they did not, but that is one scenario and not a proof.

The one frequency we measure is measured in a simulator. Section 6 is the only frequency in this paper, and its generator is ArduPilot SITL. No closure rate on real flight data is claimed anywhere. The public corpus cannot supply one: it is unannotated, so the conditioning variable does not exist in it, and reporting an unconditional closure rate as a substitute would mix hypotheses in unknown proportion and bound nothing. We have specified the real-telemetry measurement separately and have not performed it.

The governing objection, stated in full. The strongest objection to Section 5 is this one, and we reproduce it as it was put to us:

> "Unless the admissible evidence universe is defined independently, shown complete, and difficult to compute, this is a modeling observation rather than a new theorem."

We do not defeat it, and we accept all three parts. U_h is not defined independently: it is definable only relative to (M, S, R), as Section 5 now states. It is not shown complete except over a field set we enumerate and name. And it is not difficult to compute; the containment test is a linear-time subset check that runs in our test suite on every commit, and we say so affirmatively rather than letting the objection's third clause stand as an open question.

Our answer is to concede it and narrow the claim. We assert (4), which is a one-element subset check on committed source. We assert the scoped enumeration of Section 5.3, which is complete over a closed finite field set and says so. We do not assert (5) and have withdrawn it. What survives is a modeling observation about one deployed system plus measurements of what it does, and that is what this paper claims to be.

The measurement in Section 4 does not depend on defeating the objection. It is a recording of what a deployed system did under a fault, and it would read the same if U_h were never definable.

Priority is not asserted anywhere. The general mechanism of fault-state-dependent observation loss belongs to [**?**]. Section 5 states a property of one system's structure and nothing more. We assert no priority.

Simulation fidelity, and what the campaign did about it. The internal fault model produces denial by invalidating the fix flag, and real GNSS degradation is richer. Section 6 addresses this directly by adding fault classes that keep the fix, and the result is that the gate does not close for them. The remaining gap is that the ArduPilot receiver model is still a model: it dropped the fix at four satellites where a real receiver may hold one, and it reproduces neither gradual carrier-to-noise decay nor multipath.

Six flights per arm are not six independent observations. The home sites do differ, at 216 times the within-site spread, but the injection is scripted and the simulated receiver is deterministic, so the per-flight closure rate is identical across flights within each arm. The bootstrap interval is degenerate and carries no information about precision, at any n we tested. We report it as such. Wind was intended as a second axis of variation and turned out to apply no force in this build, so the campaign varied one axis and was described as varying two.

Replication remains the binding constraint. One deployed-system episode and two simulator campaigns against a single autopilot build, from 2015, is not a replication base. Every SITL number here should be repeated on a current ArduPilot and on PX4 before it is relied on, and the satellite-ladder result in particular is a statement about one simulator's GNSS model rather than about receivers.

The abstaining policy is measured, not validated. It is research code scored on three flights per arm against one simulator. Its zero nuisance cost is a property of those flights, and a policy that abstains whenever any hypothesis's support is silenced will abstain more often on a platform with more gates. Nothing in production was changed.

Two of the four exposure quantities are unavailable from operational data. T_{FH} and T_{SOFH} both condition on F_h, the ground-truth presence of the fault, which does not exist in unannotated telemetry. Only T_{NA} and T_{UH} can be estimated from a flight corpus, and the 26-log corpus used elsewhere in this project carries no health labels. Any deployment wishing to track false-healthy exposure must either annotate its corpus or run controlled injection; there is no way to recover F_h after the fact.

The escalation defect is characterised and not fixed. Section 7.1 derives it, predicts the boundary and tests the prediction. No change was made to the trust recursion, the detector decay or the failsafe dwell. A correction belongs in a separate authorised change with before-and-after evidence.

14 Claim table

Claim	Status
Under 40 s GPS denial the diagnosis reports `nominal` throughout and never names `gps_fault`	EMPIRICAL RESULT, simulated fault injection into the deployed pipeline
GPS trust reads 0.9767 over the same window	EMPIRICAL RESULT
The failsafe still reaches CAUTION via the freshness disjunct	EMPIRICAL RESULT
$\kappa(S) = 0.000$, $\kappa(S_{\text{eff}}) = 0.143$ for this system	VERIFIED FACT, computed from committed artifacts
$\Pi(S_{\text{eff}})$ is no finer than $\Pi(S)$ for this system	VERIFIED FACT by computation. The entrywise-domination *argument* for it in general is REFUTED: 483 counterexamples in 200 000 draws
$E_{\texttt{gps_fault}} \subseteq C_{\texttt{gps_fault}}$	VERIFIED FACT, one-element subset check on committed source
$U_{\texttt{gps_fault}} \subseteq C_{\texttt{gps_fault}}$ (irreparability)	NOT ESTABLISHED and not claimed. Requires a non-circular definition of the admissible evidence universe
Effective containment is complete over the currently ingested field set	VERIFIED FACT, mechanical enumeration of 6 GPS-spanning relations; formal containment is incomplete by one

Claim	Status
Effective undetectability is a threshold at $\rho = 1$	EMPIRICAL RESULT over synthesised matrices; one feature outside C_h removes absorption entirely
Chatter peaks near $1/\alpha_s$	EMPIRICAL RESULT, predicted 16.67 frames, observed 21
The repair halves $\kappa_{\max}$ with no new collision	EMPIRICAL RESULT over 1232 configurations
The design rule is checkable and enforced	VERIFIED FACT, in the test suite
Any framework in Section 2 is ours	REFUTED. All are prior work and cited as such
Detection performance on real flight data	NOT CLAIMED. Corpus is unannotated
q_{gvalid} is 0.984 under denial and 0.981 under receiver link loss	EMPIRICAL RESULT, ArduPilot SITL controlled injection, 6 flights per arm
That fourth arm measured constellation loss	CORRECTED. The satellite ladder shows the satellite count contributes nothing; the closure came from byte loss and the arm is relabelled
The gate closes at some satellite count	REFUTED. It does not close at any count down to zero, in this generator
T_{SOFH} is 23.97 of 25 s under denial on the attribution channel	EMPIRICAL RESULT, 3 flights per arm
T_{SOFH} is 0.00 s for the retained-fix glitch, whose T_{FH} is the largest of any arm at 24.64 s	EMPIRICAL RESULT. $A_h = 1$ throughout, so its exposure cannot be self-occlusion
A single exposure quantity suffices	REFUTED by our own measurement. It attributed the largest exposure to the wrong mechanism
T_{FH} on the reported status is 1.44 s	EMPIRICAL RESULT. The freshness disjunct bounds exposure; the naming failure is not bounded by it
Equation (6) predicts the shipping recursion	VERIFIED FACT, maximum absolute error 0.00 over 200 ticks; predicted crossing 93.3, observed 94
The escalation boundary $N^* + 25$ classifies the sweep	EMPIRICAL RESULT, 19 of 21 cells, both misses within 13 ticks of the line
Neither detector covers slow moderate errors	EMPIRICAL RESULT, 8 of 21 cells by detection and 9 of 21 by escalation
An abstaining state fixes the glitch exposure	REFUTED. T_{FH} unchanged at 24.64 s; the two defects need two fixes
An abstaining policy costs nuisance alerts	REFUTED on this evidence. Zero abstentions in both arms without fix loss, and no recovery penalty
No step magnitude of GNSS position error alarms	EMPIRICAL RESULT, tested to 5.6 km. A ramp of the same total error does alarm

Claim	Status
Wind varied across the WP-C6 campaign	CORRECTED. `SIM_WIND_SPD` is stored but applies no force in this build; home site did vary, at 216x the within-site spread
q_{gvalid} is 0.000 under a 62–76 m position glitch that keeps the fix	EMPIRICAL RESULT. Self-occlusion is specific to fix-loss faults
q_{gvalid} is 0.000 in healthy flight	EMPIRICAL RESULT, but the simulated receiver never loses a fix on its own, so this bounds the simulator before it bounds flight
Closure frequency of `gvalid` on real telemetry	UNMEASURED. No real logs are reachable and the corpus carries no labels
A 62–76 m position glitch raised no failsafe transition	EMPIRICAL RESULT. A separate failure mode from self-occlusion; cause not established

15 Conclusion

A deployed aerospace diagnostic layer asserted health for 96% of a fault window while the fault destroyed the evidence that would name it, with its trust channel independently agreeing. The escalation path held, through a mechanism that does not consult the diagnosis, and bounded the reported exposure to about a second. The naming failure is not bounded by anything.

The condition is narrower than it first appeared and we found that out by testing it: only faults that remove the GNSS fix produce it, and when the fix survives the layer sees the fault and declines to escalate for an unrelated reason. Neither failure is one of observability. The receiver observables that detect GNSS denial are well established, and in our own satellite ladder the fault sat in a telemetry field the receiver reported correctly and the ingestion path does not read. What we measure is evidence routing and the absence of a vocabulary for "I cannot assess this".

We claim no mathematics. The frameworks were already there; what was missing was somebody running the measurements on a real system and reporting what came back, including the parts that took claims away.

What we would tell someone building a gated diagnostic layer, all of it measured rather than argued:

1. **A hypothesis whose evidence sits entirely behind one gate needs one relation that gate does not control.** Not more evidence: one ungated relation. Absorption into the healthy class runs at 1.000 with every feature behind the gate and 0.000 with exactly one outside, at every signature density we tested. The check is a linear-time subset test over the signature matrix and the gate map, and it belongs in the build.

2. **A saturating detector with a fixed decay cannot escalate on magnitude.** Once the residual pins at zero, a larger fault buys only logarithmic extra time on an already-saturated signal. If escalation integrates evidence, the escalation condition is a duration, and it should be stated and tested as one.

3. **Size the escalation delay from the concentration cap, not from the threshold margin.** In this system the delay is dominated by the belief mass banked during calm flight, 500 units against an evidence margin worth 1. A forgetting factor chosen for smoothness silently sets the worst-case time to alarm.

4. **A monitor reading raw measurements and an estimator reading fused output have complementary blind spots, and complementary is not the same as covering.** Verify them jointly on the same fault grid. Ours are both blind to slow, moderate errors, and the uncovered region widens as the fault slows.

5. **Give the system a way to say it cannot assess something.** Absence of applicable evidence surfaced as health here because there was no other state to surface it as. Adding one removed the entire self-occlusion exposure at zero nuisance cost, and did nothing for the fault where evidence was present and ignored. Two defects, two fixes.

6. **Separate detection from escalation, and unsupported verdicts from false ones, before reporting either.** Pooling them attributed our largest measured exposure to the wrong mechanism, and we only caught it by measuring applicability separately from ground truth.